



HTML

HTML Complete Study Pack

Organized notes + fully answered interview questions

Clean reference • Study notes • Interview preparation

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Goal: Learn HTML, understand what each tag/attribute does, and have a clean reference to return to while studying or preparing for interviews.

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1. What is HTML?

HTML = HyperText Markup Language.

HTML is the standard markup language used to create web pages.

It describes the structure of a web page and consists of a series of elements.

Why learn HTML?

From the supplied session material:

- 1 Understand how the frontend works.
- 1 Build and test APIs easily.
- 1 Debug issues between frontend and backend.

2. Tags and Elements

What is an HTML Tag?

A tag is the markup used to tell the browser what an element is.

Most tags come in pairs:

```
<p>This is a paragraph.</p>
```

Here:

- 1 `<p>` = opening tag
- 1 `</p>` = closing tag
- 1 `This is a paragraph.` = content

What is an HTML Element?

An element can contain:

```
Start tag + Content + End tag
```

Example:

```
<p>Hello</p>
```

Some elements have no content and therefore have no closing tag.

Examples:

```
<br>  
<hr>  
<img>  
<input>
```

These are commonly called empty elements.

Nested Elements

A nested element is an element placed inside another element.

```
<a href="https://github.com">  
    
</a>
```

Here:

- 1 `<a>` is the parent.
- 1 `<img>` is the child.
- 1 `<img>` is nested inside `<a>`.

Do Not Duplicate an Attribute

Do not repeat the same attribute inside one tag.

```
<!-- Wrong -->

```

The supplied revision material states that the browser uses the first `src` and ignores the repeated one.

3. Basic HTML Structure

Every HTML page follows this basic structure:

```
<!DOCTYPE html>
<html>
  <head>
    <title>Page Title</title>
  </head>

  <body>
    <!-- Visible content -->
  </body>
</html>
```

Part	Purpose
<code><!DOCTYPE html></code>	Declares an HTML5 document
<code><html></code>	Root element
<code><head></code>	Metadata
<code><body></code>	Visible page content

Important

`<body>` contains visible content such as:

- 1 Headings
- 1 Paragraphs
- 1 Images
- 1 Hyperlinks
- 1 Tables
- 1 Lists

4. Head Tag

The `<head>` element is a container for metadata.

It can contain:

- 1 `<title>`
- 1 `<meta>`
- 1 Other information used by the browser/document

`<title>`

Defines the title of the document.

```
<title>My Website</title>
```

<meta>

The supplied material lists these uses:

- 1 Character set
- 1 Page description
- 1 Keywords
- 1 Author
- 1 Viewport settings

Example:

```
<meta charset="UTF-8">

<meta
  name="description"
  content="HTML study notes"
>

<meta
  name="keywords"
  content="HTML, Web Development"
>

<meta
  name="author"
  content="Student"
>

<meta
  name="viewport"
  content="width=device-width, initial-scale=1.0"
>
```

5. Text and Content

Headings — <h1> to <h6>

Headings are titles/subtitles displayed on a webpage.

```
<h1>Main Title</h1>
<h2>Section Title</h2>
<h3>Sub-section</h3>
<h4>Heading 4</h4>
<h5>Heading 5</h5>
<h6>Heading 6</h6>
```

<h1> is the highest level and <h6> is the lowest level.

SEO Note

The supplied revision sheet recommends using one <h1> as the main topic of the page.

Paragraph — <p>

Used to define a paragraph.

```
<p>This is a paragraph.</p>
```

A paragraph starts on a new line and is usually a block of text.

The browser automatically removes extra spaces and lines when the page is displayed.

Lorem Ipsum

Lorem Ipsum is placeholder text used while building a layout when the real content is not ready.

In VS Code:

```
lorem
```

then press Tab.

You can also request a number of words:

```
lorem10  
lorem50  
lorem100
```

Line Break —

Creates a new line.

```
<p>  
  Line one<br>  
  Line two  
</p>
```


 is an empty element.

Horizontal Rule — <hr>

Creates a horizontal dividing line.

```
<hr>
```

<hr> is an empty element.

Text Formatting

Tag	Purpose	Example
<code></code>	Bold	<code>Important</code>
<code><i></code>	Italic	<code><i>Emphasized</i></code>
<code><u></code>	Underline	<code><u>Underlined</u></code>
<code><mark></code>	Highlight	<code><mark>Important</mark></code>
<code></code>	Strikethrough	<code>\$100</code>
<code><sup></code>	Superscript	<code>x<sup>2</sup></code>
<code><sub></code>	Subscript	<code>H<sub>2</sub>O</code>

Examples:

```
<p>Water = H<sub>2</sub>O</p>

<p>Area = r<sup>2</sup></p>

<p>
  Old price:
  <del>$100</del>
  New price:
  <mark>$70</mark>
</p>
```

`<font>` Tag

The old `<font>` tag was used to change text color, size, and face:

```
<font color="red" size="4">
  This text is red
</font>
```

The supplied material notes that `<font>` is old-style and should be avoided in modern HTML in favor of CSS.

6. Links

Anchor — `<a>`

Creates a hyperlink.

```
<a href="https://www.google.com">
  Visit Google
</a>
```

href

Specifies the destination.

```
<a href="https://github.com">
  GitHub
</a>
```

target

Specifies where the linked document opens.

Value	Behavior
<code>_self</code>	Same tab; default
<code>_blank</code>	New tab

```
<a
  href="https://github.com"
  target="_blank"
>
  Open GitHub
</a>
```

Clickable Image

Put `<img>` inside `<a>`:

```
<a
  href="https://github.com/Eyadeltaher/Cybersecurity-notes"
  target="_blank"
>
  
</a>
```

The `<a>` is the parent and `<img>` is the child.

7. Images

`<img>`

Embeds an image.

```

```

Important Attributes

Attribute	Purpose
<code>src</code>	Image path
<code>alt</code>	Alternative text
<code>width</code>	Display width

Why is `alt` important?

The supplied revision sheet gives two main reasons:

- 1 Helps search engines understand the image.

-
- 2 Helps screen readers describe the image.

Good:

```
alt="red sports car"
```

Not useful:

```
alt="image1"
```

8. Comments

HTML comments:

```
<!-- This is a comment -->
```

Example:

```
<!-- Navigation section -->
<nav>
  <a href="#">Home</a>
</nav>

<!-- TODO: add dropdown menu here -->
```

Comments are not displayed as normal page content.

9. Layout and Containers

<div>

Defines a division or section.

It is commonly used as a block-level container for grouping elements.

```
<div style="background-color: lightblue;">
  <h2>Section Title</h2>
  <p>Content goes here.</p>
</div>
```


An inline container.

Useful for styling or targeting a small part of text.

```
<p>
  My favorite color is
  <span style="color: red;">red</span>.
</p>
```

Easy way to remember

```
<div> → block / box
<span> → inline / small part
```

10. Semantic HTML

Semantic elements clearly describe their meaning to the browser and developer.

Examples:

```
<header>
<main>
<footer>
<nav>
<section>
```

<header>

Container for introductory content.

<main>

Contains the main content of the document.

<footer>

Defines a footer for a document or section.

<nav>

Defines navigation links.

<section>

Defines a section in a document.

11. Media

Image

```

```

Video — <video>

Used to show a video.

```
<video
  src="movie.mp4"
  width="600"
  controls
  muted
  autoplay
  loop
>
  Your browser does not support video.
</video>
```

Video attributes

Attribute	Purpose
<code>controls</code>	Shows playback controls
<code>muted</code>	Starts without sound
<code>autoplay</code>	Plays automatically
<code>loop</code>	Repeats

The supplied notes mention that browsers often block autoplay with sound, so `muted + autoplay` is commonly used.

Audio — `<audio>`

Used to play audio.

```
<audio
  src="song.mp3"
  controls
  autoplay
  loop
  muted
></audio>
```

Iframe — `<iframe>`

Embeds another webpage or external content.

Examples include YouTube and Google Maps.

```
<iframe
  src="https://www.youtube.com/embed/dQw4w9WgXcQ"
  width="560"
  height="315"
>
</iframe>
```

Easy way to remember

An iframe is like a window inside your webpage that displays external content.

12. Lists

HTML has three main list types:

- 1 Ordered list
- 2 Unordered list
- 3 Description list

Ordered List — `<ol>`

Shows items in an order.

```
<ol>
  <li>HTML</li>
  <li>CSS</li>
  <li>JavaScript</li>
</ol>
```

type and start

```
<ol type="A" start="3">
  <li>Step C</li>
  <li>Step D</li>
</ol>
```

type examples:

- 1 1 numbers
- 1 A uppercase
- 1 a lowercase
- 1 I Roman numerals

start controls the starting value.

Unordered List —

Shows bullet points.

```
<ul>
  <li>HTML</li>
  <li>CSS</li>
  <li>JavaScript</li>
</ul>
```

Bullet types

Type	Style
disc	
circle	
square	

Description List — <dl>

Used for terms and their descriptions.

```
<dl>
  <dt>HTML</dt>
  <dd>The structure language of the web</dd>

  <dt>CSS</dt>
  <dd>The styling language for web pages</dd>
</dl>
```

Tag	Meaning
<dl>	Description list
<dt>	Term

Tag	Meaning
<dd>	Description

13. Tables

A table contains cells arranged in rows and columns.

```
<table border="1">
  <thead>
    <tr>
      <th>Name</th>
      <th>Age</th>
    </tr>
  </thead>

  <tbody>
    <tr>
      <td>Ahmed</td>
      <td>25</td>
    </tr>
  </tbody>

  <tfoot>
    <tr>
      <td colspan="2">End of Table</td>
    </tr>
  </tfoot>
</table>
```

Table Tags

Tag	Purpose
<table>	Table wrapper
<thead>	Header section
<tbody>	Main content
<tfoot>	Footer section
<tr>	Table row
<th>	Header cell
<td>	Data cell

Important Attributes

border

```
<table border="1">
```

Adds a border.

colspan

Merges columns horizontally.

```
<td colspan="2">Full Name</td>
```

rowspan

Merges rows vertically.

```
<td rowspan="2">Ahmed</td>
```

Memory Trick

```
colspan → columns → horizontal  
rowspan → rows → vertical
```

14. Forms

A form collects user input.

```
<form  
  action="/submit"  
  method="POST"  
>  
  <!-- inputs -->  
  <button type="submit">Send</button>  
</form>
```

A form can contain:

- 1 <input>
- 1 <label>
- 1 <select>
- 1 <textarea>
- 1 <datalist>
- 1 <button>

15. Input Types

Text

```
<input  
  type="text"  
  placeholder="Enter your name"  
  required  
>
```

Single-line text.

Password

```
<input type="password">
```

Password field.

Email

```
<input
  type="email"
  placeholder="example@mail.com"
>
```

Email input.

File

```
<input type="file">
```

File selection/upload input.

Number

```
<input
  type="number"
  min="1"
  max="100"
>
```

Numeric input.

Range

```
<input type="range">
```

Used when the exact numeric value is not important.

Search

```
<input type="search">
```

Search field.

Telephone

```
<input
  type="tel"
  placeholder="+20 100 000 0000"
>
```

Telephone number.

Why `tel` instead of `number`?

The supplied revision sheet recommends `tel` for phone numbers because phone numbers can contain:

- 1 +
- 1 spaces
- 1 leading zeros

Phone numbers should be treated as contact information rather than mathematical numbers.

Date

```
<input type="date">
```

Time

```
<input type="time">
```

Week

```
<input type="week">
```

Color

```
<input type="color">
```

Radio

Allows one selection from a group.

```
<input
  type="radio"
  name="gender"
  value="male"
>
Male

<input
  type="radio"
  name="gender"
  value="female"
>
Female
```

Checkbox

Allows zero or more selections.

```
<input
  type="checkbox"
  name="hobby"
  value="reading"
>
Reading

<input
  type="checkbox"
  name="hobby"
  value="coding"
>
Coding
```

Submit

```
<input type="submit" value="Send Form">
```

Submits form data.

Reset

```
<input type="reset">
```

Resets form values to their defaults.

16. Form/Input Attributes

placeholder

Shows a short hint inside the field.

```
<input  
  type="text"  
  placeholder="Your full name"  
>
```

The hint disappears when the user types.

required

Makes the field required.

```
<input  
  type="email"  
  required  
>
```

The form cannot be submitted while the required field is empty.

readonly

Makes the input read-only.

```
<input  
  type="text"  
  value="Read only"  
  readonly  
>
```

disabled

Disables the input.

```
<input  
  type="text"  
  value="Disabled"  
  disabled  
>
```

17. Radio and Checkbox Grouping

Radio Buttons

The important rule:

Radio buttons that belong to the same group should have the same name and different value values.

Correct:

```
<input
  type="radio"
  name="color"
  value="red"
>
Red

<input
  type="radio"
  name="color"
  value="blue"
>
Blue

<input
  type="radio"
  name="color"
  value="green"
>
Green
```

Because they have the same **name**, selecting one deselects the others.

Wrong

```
<input
  type="radio"
  name="color1"
  value="red"
>
Red

<input
  type="radio"
  name="color2"
  value="blue"
>
Blue
```

Different names make them independent.

Checkbox

Checkboxes can have the same **name** when they represent related options.

```
<input
  type="checkbox"
  name="hobby"
  value="reading"
>
Reading

<input
  type="checkbox"
  name="hobby"
  value="coding"
>
Coding
```

18. Common Attributes

Attribute	Purpose
src	Resource/image path
alt	Alternative image text
href	Link destination
target	Link opening location
width	Display width
style	Inline CSS
id	Identifier
class	Reusable class
name	Field name / radio grouping
value	Input value
type	Input/list type
placeholder	Input hint
required	Required field
readonly	Read-only field
disabled	Disabled field
action	Form submission destination
method	Form submission method
min	Minimum numeric value
max	Maximum numeric value
rows	Textarea rows
cols	Textarea width
start	Ordered-list starting value
colspan	Merge table columns
rowspan	Merge table rows
controls	Media controls
muted	No sound
autoplay	Automatic playback
loop	Repeat playback

19. Other Form Elements

<label>

Defines a label for a form element.

```
<label for="email">
  Email:
</label>

<input
  type="email"
  id="email"
>
```

<select>

Creates a dropdown list.

```
<select name="country">
  <option value="egypt">Egypt</option>
  <option value="usa">USA</option>
</select>
```

<option>

Defines one selectable option inside <select>.

<textarea>

Creates a multiline input.

```
<textarea
  rows="5"
  cols="40"
  placeholder="Write your message..."
></textarea>
```

<datalist>

Provides predefined options for an input.

```
<input list="browsers">

<datalist id="browsers">
  <option value="Chrome">
  <option value="Firefox">
  <option value="Edge">
</datalist>
```

<button>

Creates a clickable button.

```
<button type="submit">
  Send
</button>
```

20. Form Attributes

action

Defines where/what the form submission should be sent to.

```
<form action="/submit">
```

method

Specifies the HTTP method used when submitting form data.

The supplied session material lists:

1 GET

1 POST

Example:

```
<form
  action="/submit"
  method="POST"
>
```

21. id vs class

id

An **id** is intended to uniquely identify an element.

```
<div id="main-section">
  Main content
</div>
```

class

A class can be used by multiple elements.

```
<div class="card">Card 1</div>
<div class="card">Card 2</div>
```

Easy rule

```
id    → unique identifier
class → reusable group
```

22. Complete Example

```
<!DOCTYPE html>
<html lang="en">

<head>

  <meta charset="UTF-8">

  <meta
    name="description"
    content="HTML study page"
  >

  <meta
    name="keywords"
    content="HTML, Web Development"
  >

  <meta
    name="author"
    content="Student"
  >

  <meta
    name="viewport"
    content="width=device-width, initial-scale=1.0"
  >

  <title>HTML Study Page</title>

</head>

<body>

  <header>

    <h1>HTML Study Page</h1>

    <nav>
      <a href="#content">Content</a>
      <a href="#media">Media</a>
      <a href="#form">Form</a>
    </nav>
```

```
</header>

<main>

  <section id="content">

    <h2>Text</h2>

    <p>
      This is a paragraph.
    </p>

    <p>
      <b>Bold</b>
      <i>Italic</i>
      <u>Underline</u>
      <mark>Highlight</mark>
    </p>

    <p>
      H<sub>2</sub>0
      and
      x<sup>2</sup>
    </p>

    <hr>

    <h2>Links</h2>

    <a
      href="https://github.com"
      target="_blank"
    >
      Open GitHub
    </a>

    <br><br>

    <a
      href="https://github.com"
      target="_blank"
```

```
>
  
</a>

</section>

<section id="media">

  <h2>Media</h2>

  

  <br>

  <video
    src="movie.mp4"
    width="600"
    controls
    muted
    autoplay
    loop
  >
    Your browser does not support video.
  </video>

  <audio
    src="song.mp3"
    controls
  ></audio>

</section>

<section>
```

```

<h2>Lists</h2>

<h3>Ordered</h3>

<ol>
  <li>HTML</li>
  <li>CSS</li>
  <li>JavaScript</li>
</ol>

<h3>Unordered</h3>

<ul>
  <li>HTML</li>
  <li>CSS</li>
  <li>JavaScript</li>
</ul>

<h3>Description</h3>

<dl>
  <dt>HTML</dt>
  <dd>Structure of the web page</dd>

  <dt>CSS</dt>
  <dd>Styling of the web page</dd>
</dl>

</section>

<section>

  <h2>Table</h2>

  <table border="1">

    <thead>
      <tr>
        <th>Name</th>
        <th>Age</th>
      </tr>
    </thead>
  </table>

```

```
<tbody>
  <tr>
    <td>Ahmed</td>
    <td>25</td>
  </tr>
</tbody>

<tfoot>
  <tr>
    <td colspan="2">
      End of Table
    </td>
  </tr>
</tfoot>

</table>

</section>

<section id="form">

  <h2>Form</h2>

  <form
    action="/submit"
    method="POST"
  >

    <label for="name">
      Name:
    </label>

    <input
      type="text"
      id="name"
      name="name"
      placeholder="Your full name"
      required
    >

    <br><br>
```

```
<label for="email">
  Email:
</label>

<input
  type="email"
  id="email"
  name="email"
  placeholder="example@mail.com"
  required
>

<br><br>

<label for="phone">
  Phone:
</label>

<input
  type="tel"
  id="phone"
  name="phone"
  placeholder="+20 100 000 0000"
>

<br><br>

<label for="password">
  Password:
</label>

<input
  type="password"
  id="password"
  name="password"
>

<p>Gender:</p>

<input
  type="radio"
```

```
    id="male"
    name="gender"
    value="male"
  >

  <label for="male">
    Male
  </label>

  <input
    type="radio"
    id="female"
    name="gender"
    value="female"
  >

  <label for="female">
    Female
  </label>

  <p>Hobbies:</p>

  <input
    type="checkbox"
    id="reading"
    name="hobby"
    value="reading"
  >

  <label for="reading">
    Reading
  </label>

  <input
    type="checkbox"
    id="coding"
    name="hobby"
    value="coding"
  >

  <label for="coding">
    Coding
```

```
</label>

<br><br>

<label for="country">
  Country:
</label>

<select
  id="country"
  name="country"
>
  <option value="egypt">
    Egypt
  </option>

  <option value="usa">
    USA
  </option>
</select>

<br><br>

<label for="message">
  Message:
</label>

<textarea
  id="message"
  name="message"
  rows="5"
  cols="40"
  placeholder="Write your message..."
></textarea>

<br><br>

<button type="submit">
  Send
</button>

<input
```

```
        type="reset"
        value="Reset"
    >

</form>

</section>

</main>

<footer>
    <p>HTML Study Notes</p>
</footer>

</body>

</html>
```

23. Quick Revision

Structure

```
<!DOCTYPE html>
<html>
<head>
<body>
```

Text

```
<h1> → <h6>
<p>
<br>
<hr>
<b>
<i>
<u>
<mark>
<del>
<sup>
<sub>
```

Links & Images

```
<a>
<img>
```

Main attributes:

```
href
target
src
alt
width
```

Layout

```
<div>  
<span>
```

Semantic

```
<header>  
<main>  
<nav>  
<section>  
<footer>
```

Media

```
<img>  
<video>  
<audio>  
<iframe>
```

Lists

```
<ul>  
<ol>  
<dl>  
<dt>  
<dd>  
<li>
```

Tables

```
<table>  
<thead>  
<tbody>  
<tfoot>  
<tr>  
<th>  
<td>
```

Important:

```
colspan → horizontal  
rowspan → vertical
```

Forms

```
<form>  
<input>  
<label>  
<select>  
<option>  
<textarea>  
<datalist>  
<button>
```

Common inputs:

```
text
password
email
file
number
range
search
tel
date
time
week
color
radio
checkbox
submit
reset
```

Important attributes:

```
placeholder
required
readonly
disabled
name
value
type
action
method
```

Final Memory Map

```
HTML
├── Structure
│   ├── DOCTYPE
│   ├── html
│   ├── head
│   └── body
├── Text
│   ├── h1-h6
│   ├── p
│   ├── br
│   ├── hr
│   └── formatting
├── Links & Images
│   ├── a
│   └── img
├── Layout
│   ├── div
│   └── span
├── Semantic
│   ├── header
│   ├── nav
│   ├── main
│   ├── section
│   └── footer
├── Media
│   ├── video
│   ├── audio
│   └── iframe
├── Lists
│   ├── ul
│   ├── ol
│   └── dl
├── Tables
│   └── table
```



HTML Interview Questions & Answers

Every question below includes an answer. Use the question as a self-test before reading the answer.

Every question in this file has an answer. Use this file for interview preparation and revision. Read the question, answer it yourself, then open the answer.

Table of Contents

- 1 Basics
 - 1 Structure and Head
 - 1 Text and Formatting
 - 1 Links and Images
 - 1 Layout and Semantic HTML
 - 1 Media
 - 1 Lists and Tables
 - 1 Forms
 - 1 Attributes
 - 1 Rapid Revision Q&A
-

Basics

1. What is HTML?

Answer

HTML stands for HyperText Markup Language.

It is the standard markup language used to create web pages and describe their structure.

2. What is an HTML tag?

Answer

A tag is the markup syntax used to define an HTML element.

Example:

```
<p>
```

Most tags have an opening and closing tag.

3. What is an HTML element?

Answer

An HTML element can contain a start tag, content, and an end tag.

Example:

```
<p>Hello</p>
```

Some elements have no content and no closing tag, such as `<br>` and `<img>`.

4. What is the difference between a tag and an element?

Answer

A tag is the markup itself, such as `<p>`.

An element is the complete structure, such as:

```
<p>Hello</p>
```

5. What is an empty element?

Answer

An empty element does not contain content and does not have a closing tag.

Examples:

```
<br>  
<hr>  
<img>  
<input>
```

6. What is a nested element?

Answer

A nested element is an element placed inside another element.

Example:

```
<a href="https://github.com">  
    
</a>
```

Here `<img>` is nested inside `<a>`.

Structure and Head

7. What is the purpose of `<!DOCTYPE html>`?

Answer

It declares that the document is an HTML5 document.

8. What are the main parts of an HTML document?

Answer

The main parts are:

```
<!DOCTYPE html>
<html>
<head>
<body>
```

9. What is the purpose of `<html>`?

Answer

`<html>` is the root element of the HTML document and wraps the page content.

10. What is the purpose of `<head>`?

Answer

`<head>` contains metadata and document information, such as:

- 1 `<title>`
- 1 `<meta>`
- 1 character set
- 1 description
- 1 keywords
- 1 author
- 1 viewport settings

11. What is the purpose of `<body>`?

Answer

`<body>` contains the visible content of the webpage, including headings, paragraphs, images, links, tables, and lists.

12. What is the purpose of `<title>`?

Answer

`<title>` defines the title of the document/page shown by the browser.

Example:

```
<title>My Website</title>
```

13. What is <meta> used for?

Answer

The supplied session material lists these uses:

- 1 Character set
- 1 Page description
- 1 Keywords
- 1 Author
- 1 Viewport settings

Example:

```
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
```

Text and Formatting

14. What are headings in HTML?

Answer

Headings are defined from <h1> to <h6>.

```
<h1>Main Heading</h1>
<h2>Section</h2>
```

<h1> is the highest level and <h6> is the lowest.

15. What is the <p> tag?

Answer

<p> defines a paragraph.

```
<p>This is a paragraph.</p>
```

16. What is the
 tag?

Answer

 inserts a line break.

```
Line one<br>Line two
```

It is an empty element.

17. What is the `<hr>` tag?

Answer

`<hr>` creates a horizontal dividing line.

```
<hr>
```

It is an empty element.

18. What is Lorem Ipsum?

Answer

Lorem Ipsum is placeholder text used while building a layout when the real content is not ready.

In VS Code, typing:

```
lorem
```

and pressing Tab generates placeholder text.

19. What does `<b>` do?

Answer

It displays text in bold.

```
<b>Important</b>
```

20. What does `<i>` do?

Answer

It displays text in italic.

```
<i>Emphasized</i>
```

21. What does `<u>` do?

Answer

It underlines text.

```
<u>Underlined</u>
```

22. What does `<mark>` do?

Answer

It highlights text.

```
<mark>Important</mark>
```

23. What does `<del>` do?

Answer

It displays text with a strikethrough.

```
<del>$100</del>
```

24. What are `<sup>` and `<sub>`?

Answer

`<sup>` creates superscript text.

```
x<sup>2</sup>
```

`<sub>` creates subscript text.

```
H<sub>2</sub>O
```

25. What is the `<font>` tag?

Answer

The old `<font>` tag was used to change text color, size, and face.

Example:

```
<font color="red" size="4">  
  Red text  
</font>
```

The supplied notes identify it as old-style and recommend using CSS instead in modern HTML.

Links and Images

26. What is the `<a>` tag?

Answer

`<a>` creates a hyperlink.

```
<a href="https://github.com">  
  GitHub  
</a>
```

27. What is href?

Answer

`href` specifies the destination of an `<a>` link.

```
<a href="https://www.google.com">
  Google
</a>
```

28. What is target?

Answer

`target` specifies where the linked document opens.

Common values:

1 `_self` same tab

1 `_blank` new tab

Example:

```
<a
  href="https://github.com"
  target="_blank"
>
  GitHub
</a>
```

29. How can you make an image clickable?

Answer

Put the `<img>` element inside an `<a>` element.

```
<a href="https://github.com">
  
</a>
```

30. What is the `<img>` tag?

Answer

`<img>` embeds an image in a webpage.

```

```

It is an empty element.

31. What is the `src` attribute?

Answer

`src` specifies the path/source of a resource such as an image.

```

```

32. What is the alt attribute?

Answer

alt provides alternative text for an image.

It is useful when the image cannot be displayed and helps screen readers describe the image. The supplied revision material also notes its usefulness for search engines.

Example:

```

```

Layout and Semantic HTML

33. What is <div>?

Answer

<div> defines a division or section and is commonly used as a block-level container for grouping elements.

34. What is ?

Answer

 is an inline container used to style or target a small part of text.

35. What is the difference between <div> and ?

Answer

Block-level container	Inline container
Used for sections/groups	Used for small inline content
Starts a new line in normal flow	Stays within the line

Easy memory:

```
div   → box  
span  → highlighter
```

36. What is semantic HTML?

Answer

Semantic HTML uses meaningful tags that clearly describe their purpose to the browser and developer.

Examples:

```
<header>
<main>
<nav>
<section>
<footer>
```

37. What is <header>?

Answer

It represents a container for introductory content.

38. What is <main>?

Answer

It specifies the main content of the document.

39. What is <footer>?

Answer

It defines a footer for a document or section.

40. What is <nav>?

Answer

It defines navigation links.

41. What is <section>?

Answer

It defines a section in a document.

Media

42. What is <video>?

Answer

<video> is used to show a video on a webpage.

Example:

```
<video
  src="movie.mp4"
  controls
>
</video>
```

Common attributes:

1 controls
1 muted
1 autoplay
1 loop

43. What is <audio>?

Answer

<audio> is used to play an audio file.

```
<audio  
  src="song.mp3"  
  controls  
>  
</audio>
```

44. What is <iframe>?

Answer

<iframe> embeds another webpage or external content inside the current page.

Examples include YouTube and Google Maps.

45. What does controls do in video/audio?

Answer

It displays playback controls such as play/pause and volume controls.

46. What does autoplay do?

Answer

It tells the browser to start media playback automatically.

47. What does muted do?

Answer

It starts the media without sound.

48. What does loop do?

Answer

It makes the media repeat when it finishes.

Lists and Tables

49. What are the three main types of HTML lists?

Answer

- 1 Ordered list — `<ol>`
- 2 Unordered list — `<ul>`
- 3 Description list — `<dl>`

50. What is an unordered list?

Answer

`<ul>` creates a list with bullets.

```
<ul>
  <li>HTML</li>
  <li>CSS</li>
</ul>
```

51. What is an ordered list?

Answer

`<ol>` creates an ordered list.

```
<ol>
  <li>First</li>
  <li>Second</li>
</ol>
```

52. What is a description list?

Answer

A description list represents terms and their descriptions.

```
<dl>
  <dt>HTML</dt>
  <dd>Structure of the web</dd>
</dl>
```

- 1 `<dl>` list
- 1 `<dt>` term
- 1 `<dd>` description

53. What is `<li>`?

Answer

`<li>` defines a list item inside `<ul>` or `<ol>`.

54. What is <table>?

Answer

<table> creates an HTML table containing rows and columns.

55. What are <tr>, <th>, and <td>?

Answer

- 1 <tr> table row
 - 1 <th> table header cell
 - 1 <td> table data cell
-

56. What are <thead>, <tbody>, and <tfoot>?

Answer

- 1 <thead> table header section
 - 1 <tbody> main table content
 - 1 <tfoot> table footer section
-

57. What is colspan?

Answer

colspan merges table columns horizontally.

```
<td colspan="2">
  Full Name
</td>
```

58. What is rowspan?

Answer

rowspan merges table rows vertically.

```
<td rowspan="2">
  Ahmed
</td>
```

Forms

59. What is an HTML form?

Answer

<form> creates a form used to collect user input.

It can contain elements such as:

```
input
label
select
textarea
datalist
button
```

60. What is <input>?

Answer

<input> is one of the most commonly used form elements and is used to receive input from the user.

61. What are common input types?

Answer

The supplied session material includes:

```
text
password
submit
reset
color
email
file
number
range
search
tel
date
time
week
radio
checkbox
```

62. What is a radio button?

Answer

A radio button allows the user to select only one choice from a group.

```
<input
  type="radio"
  name="gender"
  value="male"
>
Male
```

63. What is a checkbox?

Answer

A checkbox allows the user to select zero or more options.

```
<input
  type="checkbox"
  name="hobby"
  value="coding"
>
Coding
```

64. How do you group radio buttons?

Answer

Give related radio buttons the same `name` and different `value` values.

```
<input
  type="radio"
  name="color"
  value="red"
>
Red

<input
  type="radio"
  name="color"
  value="blue"
>
Blue
```

The same `name` makes them one group.

65. Why should phone numbers use `type="tel"`?

Answer

The supplied revision sheet recommends `tel` because phone numbers may contain:

- 1 +
- 1 spaces
- 1 leading zeros

`tel` is intended for telephone input.

66. What does placeholder do?

Answer

It displays a short hint inside an input field.

```
<input
  type="text"
  placeholder="Your name"
>
```

The hint disappears when the user types.

67. What does required do?

Answer

It requires the user to fill in the field before submitting the form.

```
<input
  type="email"
  required
>
```

68. What does readonly do?

Answer

It makes an input field read-only.

69. What does disabled do?

Answer

It disables an input field.

70. What is <label>?

Answer

`<label>` defines a label for a form element.

```
<label for="email">
  Email:
</label>

<input
  id="email"
  type="email"
>
```

71. What is <select>?

Answer

`<select>` creates a dropdown list.

72. What is <option>?

Answer

`<option>` defines one selectable option inside a `<select>` element.

73. What is <textarea>?

Answer

`<textarea>` creates a multiline text input.

```
<textarea
  rows="5"
  cols="40"
>
</textarea>
```

74. What is <datalist>?

Answer

`<datalist>` provides predefined options for an input.

```
<input list="browsers">

<datalist id="browsers">
  <option value="Chrome">
  <option value="Firefox">
</datalist>
```

75. What is <button>?

Answer

`<button>` defines a clickable button.

```
<button type="submit">
  Send
</button>
```

Attributes

76. What are HTML attributes?

Answer

Attributes provide additional information about HTML elements.

Examples:

```
id
class
src
href
alt
style
width
target
name
value
```

77. What is id?

Answer

`id` identifies an element.

The supplied revision sheet describes `id` as unique and used for a single element.

Example:

```
<div id="main">
  Content
</div>
```

78. What is class?

Answer

`class` can be used for multiple elements.

```
<div class="card">One</div>
<div class="card">Two</div>
```

79. What is the difference between id and class?

Answer

```
id    → unique / single element
class → reusable / multiple elements
```

80. What is the style attribute?

Answer

It applies inline CSS directly to an element.

```
<p
  style="color: blue; font-size: 20px;"
>
  Styled text
</p>
```

81. What is the width attribute?

Answer

It sets the display width of supported elements such as images, tables, or videos.

```

```

82. Can you repeat the same attribute in one HTML tag?

Answer

No. The supplied revision material specifically warns against repeating the same attribute.

Wrong:

```

```

The supplied material states that the browser uses the first `src` and ignores the repeated one.

83. What is the action attribute?

Answer

`action` defines the action/destination used when the form is submitted.

```
<form action="/submit">
```

84. What is the method attribute?

Answer

`method` specifies the HTTP method used when submitting form data.

The supplied session material lists:

```
GET
POST
```

Example:

```
<form
  action="/submit"
  method="POST"
>
```

Rapid Revision Q&A

85. HTML stands for what?

Answer

HyperText Markup Language.

86. Which tag is the root of an HTML document?

Answer

`<html>`.

87. Which section contains metadata?

Answer

`<head>`.

88. Which section contains visible content?

Answer

`<body>`.

89. Which tag creates a paragraph?

Answer

`<p>`.

90. Which tag creates a hyperlink?

Answer

`<a>`.

91. Which attribute gives an `<a>` its destination?

Answer

`href`.

92. Which tag embeds an image?

Answer

`<img>`.

93. Which attribute gives alternative image text?

Answer

`alt`.

94. Which tag creates a line break?

Answer

`<br>`.

95. Which tag creates a horizontal line?

Answer

`<hr>`.

96. Which tag creates a block container?

Answer

`<div>`.

97. Which tag creates an inline container?

Answer

`<span>`.

98. Name five semantic HTML elements.

Answer

```
<header>
<nav>
<main>
<section>
<footer>
```

99. Which tag creates an unordered list?

Answer

`<ul>`.

100. Which tag creates an ordered list?

Answer

`<ol>`.

101. Which tags are used for a description list?

Answer

```
<dl>
<dt>
<dd>
```

102. Which tag creates a table?

Answer

`<table>`.

103. Which tag creates a table row?

Answer

`<tr>`.

104. Which tag creates a table header cell?

Answer

`<th>`.

105. Which tag creates a table data cell?

Answer

`<td>.`

106. What merges table columns?

Answer

`colspan.`

107. What merges table rows?

Answer

`rowspan.`

108. Which tag creates a form?

Answer

`<form>.`

109. Which tag receives user input?

Answer

`<input>.`

110. Which input type is used for email?

Answer

```
<input type="email">
```

111. Which input type is used for phone numbers?

Answer

```
<input type="tel">
```

112. Which input type allows one choice from a group?

Answer

`radio.`

113. Which input type allows multiple selections?

Answer

`checkbox`.

114. What attribute groups radio buttons?

Answer

`name`.

115. What attribute shows a hint inside an input?

Answer

`placeholder`.

116. What attribute makes a field required?

Answer

`required`.

117. What attribute makes an input read-only?

Answer

`readonly`.

118. What attribute disables an input?

Answer

`disabled`.

119. Which tag creates a dropdown?

Answer

`<select>`.

120. Which tag defines a dropdown option?

Answer

`<option>`.

121. Which tag creates multiline input?

Answer

`<textarea>`.

122. Which tag provides predefined input options?

Answer

`<datalist>`.

123. Which tag creates a clickable button?

Answer

`<button>`.

124. What does `action` do in a form?

Answer

It defines the action/destination used when the form is submitted.

125. What does `method` do in a form?

Answer

It specifies the HTTP method used to submit the form, such as [GET](#) or [POST](#).

Last-Minute Interview Cheat Sheet

HTML

→ structure of the webpage

<!DOCTYPE html>

→ HTML5 document

<html>

→ root

<head>

→ metadata

<body>

→ visible content

<h1> ... <h6>

→ headings

<p>

→ paragraph

<a>

→ link

href

→ link destination

target="_blank"

→ new tab

→ image

src

→ source/path

alt

→ alternative image text

<div>

→ block container

```
<span>
→ inline container

<header>
<main>
<nav>
<section>
<footer>
→ semantic structure

<video>
<audio>
<iframe>
→ media/external content

<ul>
→ unordered list

<ol>
→ ordered list

<dl>
→ description list

<table>
<tr>
<th>
<td>
→ tables

colspan
→ merge columns

rowspan
→ merge rows

<form>
→ user input form

<input>
→ input field
```

```
radio
→ one choice

checkbox
→ zero or more choices

name
→ groups radio buttons

placeholder
→ input hint

required
→ required field

readonly
→ read-only

disabled
→ disabled

<select>
→ dropdown

<option>
→ dropdown option

<textarea>
→ multiline input

<datalist>
→ predefined input options

<button>
→ clickable button

action
→ form submission destination

method
→ GET / POST
```

```
id  
→ unique identifier  
  
class  
→ reusable group
```